YOGESWARACHARY MODEPALLI

Data Science Engineer | Machine Learning, Python & SQL
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Portfolio | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Data science fresher with a B.Tech in Computer Science and hands-on experience building machine learning models end to end in Python, from SQL and Pandas feature engineering to model evaluation and deployment. Built a fraud detection pipeline (CatBoost + XGBoost) on 6.36M+ transaction records, ran statistical A/B tests on player-retention data, and shipped every project as a live Streamlit app. Brings 2.5 years of detail-heavy operations work and a habit of keeping data clean, documented and reproducible. Looking to apply these skills to ML and AI features in a production product team.

TECHNICAL SKILLS

Programming & Data: Python, Pandas, NumPy, SQL (joins, window functions, advanced queries), Jupyter Notebook
Machine Learning: Scikit-learn, CatBoost, XGBoost, Classification, Regression, Feature Engineering, Model Evaluation (Precision, Recall), Imbalanced Data
Statistics & Experimentation: Hypothesis Testing, A/B Testing, Exploratory Data Analysis, Forecasting
Deployment & Visualization: Streamlit, Power BI, Dashboard Development, Yahoo Finance API
Tools: Git, GitHub, VS Code, GitHub Copilot

PROJECTS

JPMorgan Chase Fraud Detection Pipeline

Python, Scikit-learn, CatBoost, XGBoost, Streamlit | GitHub | Live App | Research Paper

- Built an end-to-end fraud classification pipeline on 6.36M+ transaction records, covering feature engineering and a hybrid CatBoost/XGBoost ensemble.
- Optimised for recall on severely imbalanced data, leaving only 3 false negatives across 6.36M+ records while managing the precision trade-off.
- Deployed real-time and bulk transaction scoring as Streamlit apps and wrote an independent research paper on financial fraud patterns in India (FY 2024-25).

Cookie Cats A/B Testing & Retention Analysis

Python, Statistical Hypothesis Testing | GitHub | Live App

- Ran hypothesis tests on mobile-game retention data to compare two gate versions and measure their effect on player engagement.
- Used significance testing to turn raw event data into a clear, defensible product recommendation.

DHL Inventory Optimization

Python, Hypothesis Testing, Forecasting | GitHub | Live App

- Built a hypothesis-testing workflow (with GitHub Copilot assistance) to evaluate inventory efficiency and demand-forecasting scenarios.

ML Trading Model

Python, Regression Modeling, Yahoo Finance API | GitHub | Live App

- Built a regression-based trading signal model (with GitHub Copilot assistance) that pulls historical data from the Yahoo Finance API and falls back to synthetic data when live data is unavailable.

PDF Automation Tool (built at Gateway Visas)

Python, PyPDF2 | Internal tool, not published

- Wrote a Python tool to convert, merge, compress and edit PDFs in-house, so confidential client documents no longer had to be uploaded to third-party websites.

PROFESSIONAL EXPERIENCE

Operations and process roles; included for documentation rigor and the context behind the projects above.

Student Visa Process Executive | Gateway Visas Immigration Consultancy, Hyderabad

Aug 2024 - Sep 2025

- Cross-verified financial, academic and identity documents for 50-100 concurrent applicant cases under strict accuracy and compliance standards.
- Tracked and reported case status across a high-volume pipeline, building strong habits in structured data handling.
- Built the in-house Python PDF automation tool described above to protect client data privacy.

Process Analyst | La Global Overseas Education Consultancy, Hyderabad
May 2023 - Jul 2024

- Analysed operational and applicant-tracking data to find process bottlenecks, contributing to a 50% reduction in case turnaround time.
- Maintained audit-ready records across the full applicant lifecycle in CRM systems.

Admissions Expert | Flywise Overseas Education Consultancy, Hyderabad
Jan 2023 - Apr 2023

- Maintained applicant records and communication logs across the admissions pipeline for multiple institutional stakeholders.

Java Developer Intern | Pratian Technologies, Bengaluru
Jun 2022 - Sep 2022

- Assisted in developing and debugging Java applications in a team setting, working with Git, version control and structured testing.

EDUCATION

Bachelor of Technology, Computer Science and Engineering
Jawaharlal Nehru Technological University, Hyderabad | Jun 2017 - Nov 2020

CERTIFICATIONS

Business Analytics & Data Science - Hero Vired
Introduction to Data Science with Python - edX (HarvardX)
Generative AI for Data, Tech & Finance - Hero Vired